

Dynamical Systems of Particles Around a Kerr Black Hole

University of Toronto

Haocheng Sun

Winter 2025

- This report investigates the dynamics of particles orbiting a Kerr black hole.
- By employing Hamiltonian and Hamilton–Jacobi formalisms, we derive key conserved quantities and analyze the motion of test particles in curved spacetime.
- Numerical simulations further illustrate complex phenomena, offering insights into high-energy astrophysical processes.

Basic Settings

- Kerr Black hole has stationary and axisymmetric properties.
- We adopt the Boyer-Lindquist coordinates (t, r, θ, ϕ)
- M is the black hole mass in geometrized units ($c = G = 1$).
- spin (or rotation) parameter is defined as $a = \frac{J}{M}$ where J is the total angular momentum.

Metric of Kerr Black Hole

$$ds^2 = -\frac{\Delta - a^2 \sin^2 \theta}{\Sigma} dt^2 - 2a \sin^2 \theta \frac{r^2 + a^2 - \Delta}{\Sigma} dt d\phi \\ + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\Sigma} \sin^2 \theta d\phi^2 \quad (1)$$

where Σ and Δ are defined as

$$\Sigma = r^2 + a^2 \cos^2 \theta \quad (2)$$

$$\Delta = r^2 + a^2 - 2Mr \quad (3)$$

The Ergoregion and Horizons

- The Kerr black hole has an ergosphere where frame dragging prevents any object from remaining stationary.

$$r_{\text{ergosphere}} = M + \sqrt{M^2 - a^2 \cos^2 \theta}$$

- There are two horizons:

$$r_+ = M + \sqrt{M^2 - a^2} \quad (\text{outer event horizon})$$

$$r_- = M - \sqrt{M^2 - a^2} \quad (\text{inner Cauchy horizon})$$

- Additionally, there is a ring-shaped singularity.

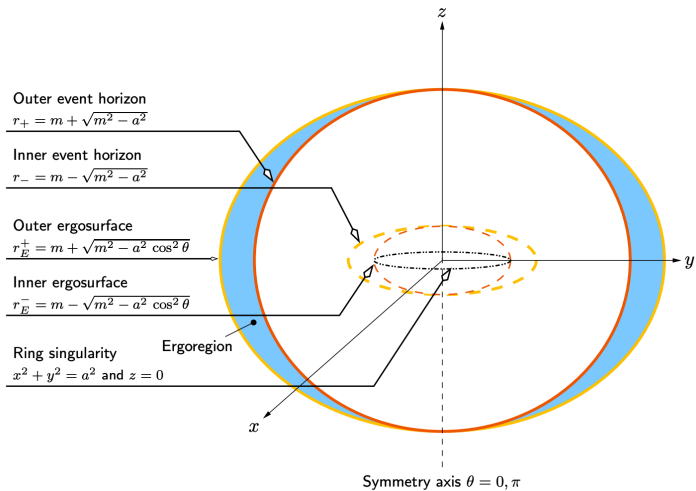


Figure: Schematic Kerr Black Hole (4)

Three Constants

As always, for a geodesic with affine parameter λ , we use x to denote the position of the geodesic, just keep in mind that we are working with Boyer-Lindquist coordinates. We use u to denote the velocity vector.

Definition

The *Lagrangian* is defined as:

$$\mathcal{L}(x, \dot{x}) := \frac{1}{2}g(\dot{x}, \dot{x}) = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu.$$

The *conjugate momentum* is given by:

$$p_\mu := \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = g_{\mu\nu} \dot{x}^\nu.$$

Notice that the physical momentum $p^\mu = u^\mu$, so lowering the index we have $p_\mu = g_{\mu\nu}u^\nu$ which is just the conjugate momentum defined above.

Lemma

p_t and p_ϕ are constant along the geodesics.

Pf: Stationary $\implies$ Killing vector $k = (1, 0, 0, 0)$.

Axisymmetric $\implies$ Killing vector $m = (0, 0, 0, 1)$ (wlog, assume rotates around the z-axis)

So consider how $\xi_\mu p^\mu$ change along the geodesic, we can show

$$p^\nu \nabla_\nu (\xi_\mu p^\mu) = 0 \quad (4)$$

- The energy and angular momentum $E = -k^\mu u_\mu - p_t$ and $L = m^\mu u_\mu p_\phi$ are constant along the geodesic.
- We should not forget there is another trivial invariant κ which is the invariant mass of the test particle, defined by

$$\kappa = g_{\mu\nu} u^\mu u^\nu = g_{\mu\nu} p^\mu p^\nu \quad (5)$$

$$\kappa = \begin{cases} -1 & \text{for timelike geodesics,} \\ +1 & \text{for spacelike geodesics,} \\ 0 & \text{for null geodesics.} \end{cases}$$

Hamilton–Jacobi Formalism

Definition

The Hamiltonian is defined as:

$$H(x, p) := p_\mu \dot{x}^\mu - \mathcal{L}(x, \dot{x}) = \frac{1}{2} p_\mu g^{\mu\nu} p_\nu.$$

Theorem

Consider the Hamiltonian-Jacobi equation

$$H\left(x, \frac{\partial S}{\partial x}\right) + \frac{\partial S}{\partial \lambda} = 0 \quad (6)$$

where $\frac{\partial S}{\partial x} = (\frac{\partial S}{\partial x^0}, \dots, \frac{\partial S}{\partial x^3})$. If $S(x(\lambda), \lambda)$ solves the Hamiltonian-Jacobi equation, then

$$\frac{\partial S}{\partial x^\mu} = p_\mu \quad \frac{\partial S}{\partial \lambda} = -H \quad (7)$$

- The solution of the Hamiltonian-Jacobi equation must have the form

$$S = -\frac{1}{2} \kappa \lambda - E t + L \phi + S^{(r\theta)}(r, \theta) \quad (8)$$

where $S^{(r\theta)}$ is a function of r and θ to be determined.

- We follow the idea of B. Carter (5), we want to find a separable solution, so we further assume S has the following expression

$$S = -\frac{1}{2} \kappa \lambda - E t + L \phi + S^{(r)}(r) + S^{(\theta)}(\theta) \quad (9)$$

Definition

$$Q := p_{\theta}^2 - \cos^2 \theta \left[(\kappa + E^2) a^2 - \frac{L^2}{\sin^2 \theta} \right].$$

The Fourth Constant: Carter Constant

Theorem

Q defined above is also constant along geodesics, it is called the Carter constant.

Pf: Substitute the above assumption of S into the H-J equation, multiply by Σ , rearrange the equation such that the LHS only involves r and the RHS only involves θ :

$$\text{LHS} = \Delta \left(\frac{dS^{(r)}}{dr} \right)^2 - \kappa r^2 - \frac{(r^2 + a^2)^2}{\Delta} E^2 + \frac{4Mra}{\Delta} EL - \frac{a^2 L^2}{\Delta} \quad (10)$$

$$\text{RHS} = - \left(\frac{dS^{(\theta)}}{d\theta} \right)^2 + \kappa a^2 \cos^2 \theta - a^2 \sin^2 \theta E^2 - \frac{1}{\sin^2 \theta} L^2 \quad (11)$$

In order to make the $\text{RHS} = -Q$, we add $a^2 E^2 + L^2$ to both sides, so the equation becomes

$$\Delta \left(\frac{dS^{(r)}}{dr} \right)^2 - \kappa r^2 - \frac{(r^2 + a^2)^2}{\Delta} E^2 + \frac{4Mra}{\Delta} EL - \frac{a^2 L^2}{\Delta} + a^2 E^2 + L^2 = -Q \quad (12)$$

Lemma

We define the functions $R(r)$ and $\Theta(\theta)$ as:

$$R(r) = \Delta \left[Q + \kappa r^2 - (L - aE)^2 \right] + \left[E(r^2 + a^2) - La \right]^2 \quad (13)$$

$$\Theta(\theta) = Q + \cos^2 \theta \left[(\kappa + E^2)a^2 - \frac{L^2}{\sin^2 \theta} \right] \quad (14)$$

Then we have the following two ODEs:

$$p_\theta^2 = (\Sigma \dot{\theta})^2 = \Theta(\theta), \quad p_r^2 = \left(\frac{\Sigma}{\Delta} \dot{r} \right)^2 = \frac{R(r)}{\Delta^2} \quad (15)$$

Pf: $p_\theta^2 = \Theta(\theta)$ follows immediately after the expression of Q . p_r is by the other expression of the Carter constant.

Theorem

the geodesics of a test particle around the Kerr black hole can be expressed as:

$$\dot{\theta} = \pm \frac{1}{\Sigma} \sqrt{\Theta}, \quad \dot{r} = \pm \frac{1}{\Sigma} \sqrt{R} \quad (16)$$

$$\dot{\phi} = -\frac{1}{\Sigma} \left(aE - \frac{L}{\sin^2 \theta} \right) + \frac{a}{\Delta \Sigma} [E(r^2 + a^2) - aL] \quad (17)$$

$$\dot{t} = -\frac{1}{\Sigma} (a^2 E \sin^2 \theta - aL) + \frac{r^2 + a^2}{\Delta \Sigma} [E(r^2 + a^2) - aL] \quad (18)$$

Pf: $\dot{\theta}$ and $\dot{r}$ is by the above lemma. The other two can be showed by solving

$$\begin{cases} p_t = g_{tt}\dot{t} + g_{t\phi}\dot{\phi} = -E \\ p_\phi = g_{\phi t}\dot{t} + g_{\phi\phi}\dot{\phi} = L \end{cases} \quad (19)$$

Numerical Solutions and Visualizations

- Although the integrals for the geodesics can be expressed in closed form, numerical methods are typically used to approximate solutions.
- The following figure shows a test particle's geodesic (red curve) around a Kerr black hole with parameters $a = 0.9$ and $M = 1$.

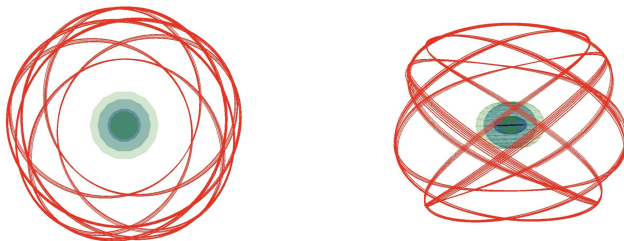
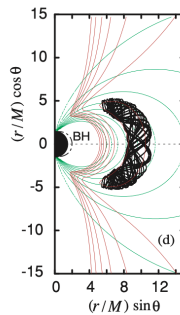
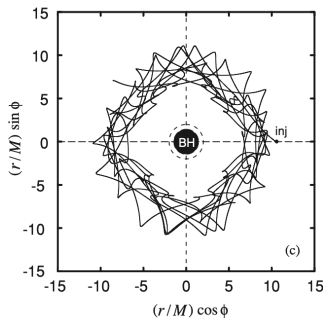
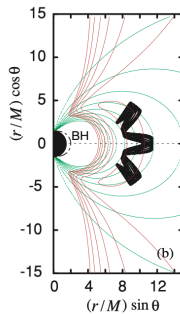
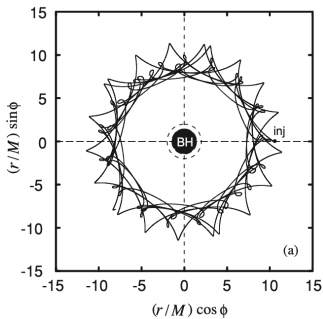


Figure: Geodesic of a test particle around a Kerr black hole (7)

Extension to Charged Particle Dynamics

- In realistic scenarios, one may need to consider charged particle dynamics in a black hole dipole magnetosphere.
- Numerical solutions of the equations of motion show complex behavior such as gyration around magnetic field lines, toroidal drifts, and multi-periodic motion in the poloidal plane.
- Unlike the neutral test particle case, an additional fourth constant of motion is lost when the electromagnetic field is present, and the motion can become non-integrable (chaotic).
- The picture below are trajectories of a test charge for $a = 0.9M$ with different initial conditions.



- [1] B. O'Neill, *The Geometry of Kerr Black Holes*, A K Peters/CRC Press, 1995.
- [2] B. Schutz, *A First Course in General Relativity*, 2nd ed., Cambridge University Press, 2009.
- [3] R. M. Wald, *General Relativity*, The University of Chicago Press, 1984.
- [4] M. Visser, *The Kerr Spacetime: A Brief Introduction*, arXiv:0706.0622 [gr-qc], 2007.
- [5] B. Carter, *Global Structure of the Kerr Family of Gravitational Fields*, Phys. Rev. **174**, 1559 (1968).
- [6] H. Goldstein, *Classical Mechanics*, 3rd ed., Addison Wesley, 2001, Chapter 10.
- [7] A. Nieddu, *Kerr Black Holes*,
<https://alexnieddu.github.io/Kerr-Black-Holes/>, (Accessed: March 24, 2025).
- [8] M. Takahashi and H. Koyama, *Chaotic motion of Charged Particles in an Electromagnetic Field Surrounding a Rotating Black Hole*, arXiv:0807.0277v1 [gr-qc], 2008.